4.6

PAEDIATRICS

Febrile Neutropenia

CHAPTER 4.6	FEBRILE NEUTROPENIA		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Non-neutropenic fever			
Stable patient, no previous infection, no infection foci			
S. epidermidis (MSSE) *S. aureus* (MSSA) *Streptococcus viridans* Enterobacteriaceae Pseudomonas Enterococcus	Cloxacillin 50 mg/kg/ dose (max 2 g) IV q6H + Gentamicin 5 mg/kg/ dose (max 240 mg) IV q24H	Ceftriaxone 50 mg/kg/ dose IV q12H	
Suspected in-dwelling venous catheter as source			

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Gram-negative bacilli *S. aureus* (MRSA) *S. epidermidis* (MRSE)	Ceftazidime 50 mg/kg (max 2 g) IV q8H + Vancomycin 15 mg/kg (max 1 g) IV q8H		• Monitor Vancomycin concentration via therapeutic drug monitoring and renal function.

Common causative organisms for febrile neutropenia in paediatrics population		
Gram-positive organisms	**Gram-negative organisms**	**Fungal organisms**
S. aureus (MSSA, MRSA) *Streptococcus spp.* *Enterococcus spp.*	*Escherichia coli* *Klebsiella pneumoniae* *Pseudomonas aeruginosa* *Enterobacter spp.* *Acinetobacter baumannii* *Stenotrophomonas maltophilia* *Citrobacter spp.*	*Candida spp.* *Aspergillus spp.*

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Febrile neutropenia			
Empirical treatment			
Stable patient	Piperacillin/Tazobactam 100 mg/kg/dose (Piperacillin component) (max 4 g) IV q8H	Meropenem 20-40 mg/kg/dose (max 2 g) IV q8H	
HSCT patient	Cefepime 50 mg/kg/dose (max 2 g) IV q8H		
Unstable patient	Meropenem 20-40 mg/kg/dose (max 2 g) IV q8H		
History of previous MRSA colonisation/ clinically suspected line-related infection	As **ANY** of the above and **ADD** Vancomycin 15 mg/kg/dose (max 1 g) IV q8H		
Documented fungemia **OR**	As **ANY** of the above and **ADD**		• Please monitor renal function. • Amphotericin B infusion should be administered over 4 to 6 hours.

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
No clinical improvement/ persistently febrile after 96 hours of antibiotics	Fluconazole 6-12 mg/kg/dose IV q24H **OR** Amphotericin B Deoxycholate 1 mg/kg IV q24H*		
Directed treatment			
Pneumocystis jirovecii pneumonia (PJP/PCP)	Co-trimoxazole 5 mg/kg/dose (Trimethoprim component) (max 160 mg) IV q6H	Pentamidine* 4 mg/kg/dose IV q24H	• Duration of treatment: 21 days (may convert IV to PO when feasible). • Moderate to severe: Consider Prednisolone 1 - 2 mg/kg/day (max 80 mg) q12H in the first week; then taper over the next 2 weeks. • ***Pentamidine must be administered slowly over 1 hour in D5W to avoid risk of severe hypotension.**
Stenotrophomonas maltophilia	Co-trimoxazole 4-5 mg/kg/dose (Trimethoprim component) IV q6-8H	**Older children:** Levofloxacin 8-10 mg/kg/ dose IV q12H	• Avoid the Fluoroquinolone group, if possible, for patients < 18 years old. Risk of ruptured tendo-achilles and arthropathies.
Candida spp.			• Consider removal of CVC.

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
(Other than *C. glabrata and C. krusei)* **Stable haemodynamic** **Unstable haemodynamic, recent azoles exposure, unknown *Candida* spp.**	Fluconazole 6-12 mg/kg/ dose IV / PO q24H **OR** Amphotericin B Deoxycholate 1 mg/kg/dose IV q24H Micafungin 2 mg/kg/dose (max 100 mg) IV q24H **OR** Amphotericin B Lipid Complex 5 mg/kg/dose IV q24H		• Repeat blood culture every 72 hours till a negative culture. • Duration of treatment: ≥ 21 days on average or for 2 weeks following first negative blood culture. • Liposomal Amphotericin B is not more effective than Amphotericin B as an antifungal.
Aspergillus spp.	Voriconazole 9 mg/kg/dose (max 350 mg) IV q12H for 2 doses **then followed by** 8 mg/kg/dose IV q12H	Amphotericin B Deoxycholate 1-1.5 mg/kg/dose IV q24H	• Fluconazole has no activity against *Aspergillus* spp. • Please monitor renal function.

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<tr><th>CHAPTER 3.6</th><th colspan="3">FEBRILE NEUTROPENIA</th></tr>
<tr><th rowspan="2">Infections and Likely Organisms</th><th colspan="2">Suggested Treatment</th><th rowspan="2">Notes</th></tr>
<tr><th>Recommended</th><th>Alternative</th></tr>
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<td>Varicella zoster

Post-exposure prophylaxis

Effective up to 10 days following exposure.

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Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Cytomegalovirus (CMV)	Ganciclovir 5 mg/kg/dose IV q12H		• Duration of treatment: 14 – 21 days.